

Table 1
OUTDOOR DESIGN CONDITIONS FOR UNITED STATES AND CANADA
DESIGN GRAINS BASED ON AN INSIDE DESIGN TEMPERATURE OF 75°F

Location	Latitude Degrees	WINTER			SUMMER			
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
ALABAMA								
Alexander City	33	22		93	76	37	44	21 M
Anniston AP	33	22	2810	94	76	35	42	21 M
Auburn	32	22		93	76	37	44	21 M
Birmingham AP	33	21	2710	94	75	30	37	21 M
Decatur	34	16	3050	93	74	25	32	22 M
Dothan AP	31	27	1400	92	76	39	46	20 M
Florence AP	34	21	3199	94	74	23	30	22 M
Gadsden	34	20	3000	94	75	30	37	22 M
Huntsville AP	34	16	3190	93	74	25	33	23 M
Mobile AP	30	29	1620	93	77	44	51	18 M
Mobile CO	30	29	1620	93	77	44	51	16 M
Montgomery AP	32	25	2250	95	76	33	40	21 M
Selma-Craig AFB	32	26	2160	95	77	38	47	21 M
Talladega	33	22		94	76	33	42	21 M
Tuscaloosa AP	33	23	2590	96	76	32	39	22 M
ALASKA								
Anchorage AP	61	-18	10860	68	58	0	0	15 L
Barrow (S)	71	-41	20265	53	50	0	0	12 L
Fairbanks AP (S)	64	-47	14290	78	60	0	0	24 M
Juneau AP	58	1	9080	70	58	0	0	15 L
Kodiak	57	13	8860	65	56	0	0	10 L
Nome AP	64	-27	14170	62	55	0	0	10 L
ARIZONA								
Douglas AP	31	31	2630	95	63	0	0	31 H
Flagstaff AP	35	4	7290	82	55	0	0	31 H
Fort Huachuca AP (S)	31	28	2551	92	62	0	0	27 H
Kingman AP	35	25		100	64	0	0	30 H
Nogales	31	32	2150	96	64	0	0	31 H
Phoenix AP (S)	33	34	1680	107	71	0	0	27 H
Prescott AP	34	9		94	60	0	0	30 H
Tuscon AP (S)	32	32	1700	102	66	0	0	26 H
Winslow AP	35	10	4780	95	60	0	0	32 H
Yuma AP	32	39	970	109	72	0	0	27 H
ARKANSAS								
Blytheville AFB	36	15	3760	94	77	42	49	21 M
Camden	33	23		96	76	32	39	21 M
El Dorado AP	33	23	2300	96	76	32	39	21 M
Fayetteville AP	36	12	3840	94	73	18	25	23 M
Fort Smith AP	35	17	3290	98	76	29	36	24 M
Hot Springs	34	23	2729	97	77	37	44	22 M
Jonesboro	35	15		94	77	42	49	21 M
Little Rock AP (S)	34	20	3170	96	77	39	46	22 M
Pine Bluff AP	34	22	2588	97	77	37	44	22 M
Texarkana AP	33	23	2530	96	77	39	46	21 M
CALIFORNIA								
Bakersfield AP	35	32	2150	101	69	0	0	32 H
Barstow	34	29	2203	104	68	0	0	37 H
Blythe AP	33	33		110	71	0	0	28 H
Burbank AP	34	39	1700	91	68	0	2	25 M
Chico	39	30	2835	101	68	0	0	36 H
Concord	38	27	3035	97	68	0	0	32 H
Covina	34	35		95	68	0	0	31 H
Crescent City AP	41	33		65	59	0	0	18 M
Downey	34	40		89	70	9	16	22 M
El Cajon	32	44		80	69	17	24	30 H
El Centro AP (S)	32	38	925	110	74	0	4	34 H
Escondido	33	41		85	68	4	11	30 H
Eureka/Arcata AP	41	33	4640	65	59	0	0	11 L
Fairfield-Travis AFB	38	32	2725	95	67	0	0	34 H
Fresno AP (S)	36	30	2610	100	69	0	0	34 H
Hamilton AFB	38	32	3311	84	66	0	3	28 H
Laguna Beach	33	43		80	68	13	20	18 M
Livermore	37	27	3035	97	68	0	0	24 M
Lompoc, Vandenberg AFB	34	38	3451	70	61	0	0	20 M
Long Beach AP	33	43	1803	80	68	13	20	22 M
Los Angeles AP (S)	34	43	1960	80	68	13	20	15 L
Los Angeles CO (S)	34	40	1960	89	70	9	16	20 M
Merced-Castle AFB	37	31	2470	99	69	0	0	36 H
Modesto	37	30		98	68	0	0	36 H
Monterey	36	38	2750	71	61	0	0	20 M

AP - Airport
CO - City Office
S - Solar Data Available

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Napa	38	32		96	68	0	0	30 H
Needles AP	34	33		110	71	0	0	27 H
Oakland AP	37	36	2940	80	63	0	0	19 M
Oceanside	33	43		80	68	13	20	13 L
Ontario	34	33	2009	99	69	0	5	36 H
Oxnard	34	36		80	64	0	0	19 M
Palmdale AP	34	22	2929	101	65	0	0	35 H
Palm Springs	33	35		110	70	0	0	35 H
Pasadena	34	35	1694	95	68	0	0	29 H
Petaluma	38	29		90	66	0	0	31 H
Pomona CO	34	30	2166	99	69	0	0	36 H
Redding AP	40	31		102	67	0	0	32 H
Redlands	34	33		99	69	0	0	33 H
Richmond	38	36		80	63	0	0	17 M
Riverside-March AFB (S)	33	32	2162	98	68	0	0	37 H
Sacramento AP	38	32	2700	98	70	0	2	36 H
Salinas AP	36	32		70	60	0	0	24 M
San Bernadino, Norton AFB	34	33	1978	99	69	0	0	38 H
San Diego AP	32	44	1500	80	69	17	24	12 L
San Fernando	34	39		91	68	0	2	38 H
San Francisco AP	37	38	3040	77	63	0	0	20 M
San Francisco CO	37	40	3040	71	62	0	3	14 L
San Jose AP	37	36	2416	81	65	0	3	26 H
San Luis Obispo	35	35	2472	88	70	10	17	26 H
Santa Ana AP	33	39	1675	85	68	4	11	28 H
Santa Barbara MAP	34	36	2470	77	66	7	14	24 M
Santa Cruz	37	38		71	61	0	0	28 H
Santa Maria AP (S)	34	33	2930	76	63	0	2	23 M
Santa Monica CO	34	43		80	68	13	20	16 M
Santa Paula	34	35		86	67	0	4	36 H
Santa Rosa	38	29	3065	95	67	0	0	34 H
Stockton AP	37	30	2806	97	68	0	0	37 H
Ukiah	39	29		95	68	0	0	40 H
Visalia	36	30		100	69	0	0	38 H
Yreka	41	17		92	64	0	0	38 H
Yuba City	39	31		101	67	0	0	36 H
COLORADO								
Alamosa AP	37	-6	8529	82	57	0	0	35 H
Boulder	40	0		91	59	0	0	27 H
Colorado Springs AP	38	2	6410	88	57	0	0	30 H
Denver AP	39	1	6150	91	59	0	0	28 H
Durango	37	-1		87	59	0	0	30 H
Fort Collins	40	1		91	59	0	0	28 H
Grand Junction AP (S)	39	7	5660	94	59	0	0	29 H
Greeley	40	4		94	60	0	0	29 H
LaJunta AP	38	3	5132	98	68	0	0	31 H
Leadville	39	-14		81	51	0	0	30 H
Pueblo AP	38	0	5480	95	61	0	0	31 H
Sterling	40	-2		93	62	0	0	30 H
Trinidad AP	37	3	5330	91	61	0	0	32 H
CONNECTICUT								
Bridgeport AP	41	9	5617	84	71	22	29	18 M
Hartford, Brainard Field	41	7	6170	88	73	28	35	22 M
New Haven AP	41	7	5890	84	73	33	40	17 M
New London	41	9		85	72	26	33	16 M
Norwalk	41	9		84	71	22	29	19 M
Norwich	41	7		86	73	31	38	18 M
Waterbury	41	2	6672	85	71	20	27	21 M
Windsor Locks, Bradley Field (S)	42	4	6350	88	72	22	29	22 M
DELAWARE								
Dover AFB	39	15	4700	90	75	36	43	18 M
Wilmington AP	39	14	4930	89	74	41	38	20 M
DISTRICT OF COLUMBIA								
Andrews AFB	38	14		90	74	30	37	18 M
Washington National AP	38	17	4240	91	74	29	36	18 M
FLORIDA								
Belle Glade	26	44		91	76	41	48	16 M

AP - Airport
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Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Cape Kennedy AP	28	38	711	88	78	59	66	15 L
Daytona Beach AP	29	35	879	90	77	49	56	15 L
Fort Lauderdale	26	46	244	91	78	53	60	15 L
Fort Myers AP	26	44	442	92	78	51	58	18 M
Fort Pierce	27	42		90	78	55	62	15 L
Gainesville AP (S)	29	31	1081	93	77	45	51	18 M
Jacksonville AP	30	32	1230	94	77	42	49	19 M
Key West AP	24	57	110	90	78	55	62	9 L
Lakeland CO (S)	28	41	661	91	76	51	48	17 M
Miami AP (S)	25	47	200	90	77	49	56	15 L
Miami Beach CO	25	48	200	89	77	51	58	10 L
Ocala	29	34		93	77	44	51	18 M
Orlando AP	28	38	720	93	76	37	44	17 M
Panama City, Tyndall AFB	30	33	1390	90	77	49	56	14 L
Pensacola CO	30	29	1480	93	77	44	51	14 L
St. Augustine	29	35	1051	89	78	57	64	16 M
St. Petersburg	28	40	670	91	77	47	54	16 M
Sanford	28	38		93	76	37	44	17 M
Sarasota	27	42		92	77	45	52	17 M
Tallahassee AP (S)	30	30	1520	92	76	39	46	19 M
Tampa AP (S)	28	40	700	91	77	47	54	17 M
West Palm Beach AP	26	45	270	91	78	53	60	16 M
GEORGIA								
Albany, Turner AFB	31	29	1760	95	76	33	40	20 M
Americus	32	25		94	76	35	42	20 M
Athens	34	22	2929	92	74	27	34	21 M
Atlanta AP (S)	33	22	2990	92	74	27	34	19 M
Augusta AP	33	23	2410	95	76	33	40	19 M
Brunswick	31	32	1531	89	78	57	64	18 M
Columbus, Lawson AFB	32	24	2380	93	76	37	44	21 M
Dalton	34	22		93	76	37	44	22 M
Dublin	32	25		93	76	37	44	20 M
Gainesville	34	21		91	74	29	36	21 M
Griffin (S)	33	22		90	75	36	43	21 M
La Grange	33	23		91	75	35	42	21 M
Macon AP	32	25	2160	93	76	37	44	22 M
Marietta, Dobbins AFB	34	21	3080	92	74	27	34	21 M
Moultrie	31	30		95	77	40	47	20 M
Rome AP	34	22	3326	93	76	37	44	23 M
Savannah-Travis AP	32	27	1850	93	77	44	51	20 M
Valdosta-Moody AFB	31	31	1520	94	77	43	49	20 M
Waycross	31	29		94	77	43	49	20 M
HAWAII								
Hilo AP (S)	19	62	0	83	72	30	37	15 L
Honolulu AP	21	63	0	86	73	31	38	12 L
Kaneohe Bay MCAS	21	66	0	84	74	40	47	12 L
Wahaiwa	21	59	10	85	72	26	33	14 L
IDAHO								
Boise AP (S)	43	10	5830	94	64	0	0	31 H
Burley	42	2		95	61	0	0	35 H
Coeur D'Alene AP	47	-1	6660	86	61	0	0	31 H
Idaho Falls AP	43	-6	7890	87	61	0	0	38 H
Lewiston AP	46	6	5500	93	64	0	0	32 H
Moscow	46	0		87	62	0	0	32 H
Mountain Home AFB	43	12	6120	97	63	0	0	36 H
Pocatello AP	43	-1	7030	91	60	0	0	35 H
Twin Falls AP (S)	42	2	6730	95	61	0	0	34 H
ILLINOIS								
Aurora	41	-1	6660	91	76	41	48	20 M
Belleville, Scott AFB	38	6	4480	92	76	39	49	21 M
Bloomington	40	-2		90	74	30	37	21 M
Carbondale	37	7		93	77	44	51	21 M
Champaign/Urbana	40	2	5800	92	74	27	34	21 M
Chicago, Midway AP	41	0	6160	91	73	23	29	20 M
Chicago, O'Hare AP	42	-4	6640	89	74	31	38	20 M
Chicago CO	41	2	6640	91	74	29	36	15 L
Danville	40	1	5538	90	74	30	37	21 M
Decatur	39	2	5480	91	74	29	36	21 M
Dixon	41	-2		90	74	30	37	23 M
Elgin	42	-2		88	74	33	40	21 M
Freeport	42	-4		89	73	26	33	24 M

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Galesburg	41	-2	6005	91	75	35	42	22 M
Greenville	39	4		92	75	33	40	21 M
Joliet	41	0	6180	90	74	30	37	20 M
Kankakee	41	1	6040	90	74	30	37	21 M
LaSalle/Peru	41	-2		91	75	35	42	22 M
Macomb	40	0		92	76	39	46	22 M
Moline AP	41	-4	6410	91	75	35	42	23 M
Mt. Vernon	38	5		92	75	33	40	21 M
Peoria AP	40	-4	6070	89	74	31	38	22 M
Quincy AP	40	3	5267	93	76	37	44	22 M
Rantoul, Chanute AFB	40	1	5966	91	74	29	36	21 M
Rockford	42	-4	6840	89	73	26	33	24 M
Springfield AP	39	2	5530	92	74	27	34	21 M
Waukegan	42	-3		89	74	31	37	21 M
INDIANA								
Anderson	40	6	5580	92	75	33	40	22 M
Bedford	38	5		92	75	33	40	22 M
Bloomington	39	5	4860	92	75	33	40	22 M
Columbus, Bakalar AFB	39	7	5132	92	75	33	40	22 M
Crawfordsville	40	3		91	74	29	36	22 M
Evansville AP	38	9	4500	93	75	31	38	22 M
Fort Wayne AP	41	1	6220	89	72	20	27	24 M
Goshen AP	41	1		89	73	26	33	23 M
Hobart	41	2		88	73	26	33	21 M
Huntington	40	1		89	72	20	27	23 M
Indianapolis AP (S)	39	2	5630	90	74	30	37	22 M
Jeffersonville	38	10		93	74	25	33	23 M
Kokoma	40	0	5590	90	73	24	31	22 M
Lafayette	40	3	5820	91	73	23	30	22 M
LaPorte	41	3		90	74	30	37	22 M
Marion	40	0		90	73	24	31	23 M
Muncie	40	2	5950	90	73	24	31	22 M
Peru, Bunker Hill AFB	40	-1		88	73	28	35	22 M
Richmond AP	39	2		90	74	30	37	22 M
Shelbyville	39	3		91	74	29	36	22 M
South Bend AP	41	1	6460	89	73	26	33	22 M
Terre Haute AP	39	4	5360	92	74	27	34	22 M
Valparaise	41	3		90	74	30	37	22 M
Vincennes	38	6		92	74	27	34	22 M
IOWA								
Ames (S)	42	-6		90	74	30	37	23 M
Burlington AP	40	-3	6120	91	75	35	43	22 M
Cedar Rapids AP	41	-5	6600	88	75	38	45	23 M
Clinton	41	-3		90	75	36	33	23 M
Council Bluffs	41	-3	6610	91	75	35	42	22 M
Des Moines AP	41	-5	6610	91	74	29	36	23 M
Debuque	42	-7	7380	88	73	28	35	22 M
Fort Dodge	42	-7	7070	88	74	33	40	23 M
Iowa City	41	-6	6404	89	76	44	51	22 M
Keokuk	40	0	6404	92	75	33	40	22 M
Marshalltown	42	-7	6850	90	75	36	43	23 M
Mason City AP	43	-11	7790	88	74	33	40	24 M
Newton	41	-5		91	74	29	36	23 M
Ottumwa AP	41	-4		91	74	29	36	22 M
Sioux City AP	42	-7	6960	92	74	27	34	24 M
Waterloo	42	-10	7370	89	75	38	45	23 M
KANSAS								
Atchison	39	2		93	76	37	44	23 M
Chanute AP	34	7	4566	97	74	19	26	23 M
Dodge City AP (S)	37	5		97	69	0	0	25 M
El Dorado	37	7	4990	98	73	11	18	24 M
Emporia	38	5		97	74	19	26	25 M
Garden City AP	38	4		96	69	0	0	28 H
Goodland AP	39	0	6140	96	65	0	0	31 H
Great Bend	38	4		98	73	11	18	28 H
Hutchinson AP	38	8	4680	99	72	3	10	28 H
Liberal	37	7		96	68	0	0	28 H
Manhattan, Fort Riley (S)	39	3	5306	95	75	28	35	24 M
Parsons	37	9	4158	97	74	19	26	23 M
Russel AP	38	4		98	73	11	18	29 H
Salina	38	5	4970	100	74	13	20	26 H
Topeka AP	39	4	5210	96	75	26	33	24 M

AP - Airport
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Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½ % Design db	Heating D.O. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Wichita AP	37	7	4640	98	73	11	18	23 M
KENTUCKY								
Ashland	38	10	4555	91	74	29	36	22 M
Bowling Green AP	37	10	4280	92	75	33	40	21 M
Corbin AP	37	9		92	73	21	28	23 M
Covington AP	39	6	5260	90	72	19	26	22 M
Hopkinsville, Campbell AFB	36	10	4290	92	75	33	40	21 M
Lexington AP (S)	38	8	4760	91	73	23	30	22 M
Louisville AP	38	10	4610	93	74	25	33	23 M
Madisonville	37	10		93	75	31	38	22 M
Owensboro	37	10	4200	94	75	30	37	23 M
Paducah	37	12	3650	95	75	28	35	20 M
LOUISIANA								
Alexandria AP	31	27	2000	94	77	43	49	20 M
Baton Rouge AP	30	29	1610	93	77	44	51	19 M
Bogalusa	30	28		93	77	44	51	19 M
Houma	29	35		93	78	50	57	15 L
Lafayette AP	30	30	1550	94	78	49	56	18 M
Lake Charles AP (S)	30	31	1490	93	77	44	51	17 M
Minden	32	25		96	76	32	38	20 M
Monroe AP	32	25	2310	96	76	32	38	20 M
Natchitoches	31	26		95	77	40	47	20 M
New Orleans AP	30	33	1400	92	78	51	58	16 M
Shreveport AP (S)	32	25	2160	96	76	32	39	20 M
MAINE								
Augusta AP	44	-3	7826	85	70	15	23	22 M
Bangor, Dow AFB	44	-6	8220	83	68	8	15	22 M
Caribou AP (S)	46	-13	9770	81	67	7	14	21 M
Lewiston	44	-2	7690	85	70	15	22	22 M
Millinocket AP	45	-9	8533	83	68	8	15	22 M
Portland (S)	43	-1	7570	84	71	23	29	22 M
Waterville	44	-4		84	69	11	18	22 M
MARYLAND								
Baltimore AP	39	13	4680	91	75	35	42	21 M
Baltimore CO	39	17		89	76	43	51	17 M
Cumberland	39	10	5070	89	74	31	38	22 M
Frederick AP	39	12	5030	91	75	35	42	22 M
Hagerstown	39	12	5130	91	74	29	36	22 M
Salisbury (S)	38	16	4220	91	75	35	42	18 M
MASSACHUSETTS								
Boston AP (S)	42	9	5630	88	71	16	23	16 M
Clinton	42	2		87	71	18	25	17 M
Fall River	41	9	5774	84	71	22	29	18 M
Framingham	42	6		86	71	19	26	17 M
Gloucester	42	5		86	71	19	26	15 L
Greenfield	42	-2		85	71	20	27	23 M
Lawrence	42	0	6195	87	72	23	30	22 M
Lowell	42	1	6060	88	72	22	29	21 M
New Bedford	41	9	5400	82	71	26	33	19 M
Pittsfield AP	42	-3	7580	84	70	16	23	23 M
Springfield, Westover AFB	42	0	5840	87	71	18	25	19 M
Taunton	41	9		86	72	25	32	18 M
Worcester AP	42	4	6970	84	70	16	23	18 M
MICHIGAN								
Adrian	41	3		88	72	22	29	23 M
Alpena AP	45	-6	8510	85	70	15	22	27 H
Battle Creek AP	42	5	6580	88	72	22	29	23 M
Benton Harbor AP	42	5		88	72	22	29	20 M
Detroit	42	6	6290	88	72	22	29	20 M
Escanaba	45	-7	8481	83	69	13	20	17 M
Flint AP	42	1	7200	87	72	23	30	25 M
Grand Rapids AP	42	5	6890	88	72	22	29	24 M
Holland	42	6		86	71	19	26	22 M
Jackson AP	42	5		88	72	22	29	23 M
Kalamazoo	42	5		88	72	22	29	23 M
Lansing AP	42	1	6940	87	72	23	30	24 M
Marquette CO	46	-8	8390	81	69	16	23	18 M

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER		Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb			
Mt. Pleasant	43	4		87	72	23	30	24 M
Muskegon AP	43	6	6700	84	70	16	23	21 H
Pontiac	42	4		87	72	23	30	21 M
Port Huron	43	4	6564	87	72	23	30	21 H
Saginaw AP	43	4	7120	87	72	23	30	23 M
Sault								
Ste. Marie AP (S)	46	-8	9050	81	69	16	23	23 M
Traverse City AP	44	1	7700	86	71	19	26	22 M
Ypsilanti	42	5	6424	89	71	15	22	22 M
MINNESOTA								
Albert Lea	43	-12		87	72	23	30	24 M
Alexandria AP	45	-16		88	72	22	29	24 M
Bemidji AP	47	-26	10203	85	69	9	16	24 M
Brainerd	46	-16		87	71	18	25	24 M
Duluth AP	46	-16	9890	82	68	10	17	22 M
Fairbault	44	-12		88	72	22	27	24 M
Fergus Falls	46	-17		88	72	22	27	24 M
International Falls AP	48	-25	10600	83	68	8	15	26 M
Mankato	44	-12	8310	88	72	22	29	24 M
Minneapolis/St. Paul AP	44	-12	8250	89	73	26	33	22 M
Rochester AP	44	-12	8295	87	72	23	30	24 M
St. Cloud AP (S)	45	-11	8890	88	72	22	29	24 M
Virginia	47	-21		83	68	8	15	23 M
Wilmar	45	-11		88	72	22	29	24 M
Winona	44	-10		88	73	28	35	24 M
MISSISSIPPI								
Biloxi, Keesler AFB	30	31	1500	92	79	58	65	16 M
Clarksdale	34	19		94	77	42	49	21 M
Columbus AFB	33	20	2890	93	77	44	51	22 M
Greenville AFB	33	20	2580	93	77	44	51	21 M
Greenwood	33	20		93	77	44	51	21 M
Hattiesburg	31	27	1840	94	77	42	49	21 M
Jackson AP	32	25	2260	95	76	33	40	21 M
Laurel	31	27		94	77	42	49	21 M
McComb AP	31	26		94	76	35	42	18 M
Meridian AP	32	23	2340	95	76	33	40	22 M
Natchez	31	27	1800	94	78	49	56	21 M
Tupelo	34	19		94	77	42	49	22 M
Vicksburg CO	32	26	2040	95	78	47	54	21 M
MISSOURI								
Cape Girardeau	37	13		95	75	28	35	21 M
Columbia AP (S)	39	4	5070	94	74	23	30	22 M
Farmington AP	37	8		93	75	31	38	22 M
Hannibal	39	3	5512	93	76	37	44	22 M
Jefferson City	38	7	4620	95	74	21	28	23 M
Joplin AP	37	10	4090	97	73	13	20	24 M
Kansas City AP	39	6	4750	96	74	20	27	20 M
Kirksville AP	40	0		93	74	25	32	24 M
Mexico	39	4		94	74	23	30	22 M
Moberly	39	3		94	74	23	30	23 M
Poplar Bluff	36	16	3910	95	76	32	40	22 M
Rolla	38	9		91	75	35	42	22 M
St. Joseph AP	39	2	5440	93	76	37	44	23 M
St. Louis AP	38	6	4900	94	75	30	37	21 M
St. Louis CO	38	8		94	75	30	37	18 M
Sedalia, Whiteman AFB	38	4	5012	92	76	39	46	22 M
Sikeston	36	15		95	76	33	40	21 M
Springfield AP	37	9	4900	93	74	25	32	23 M
MONTANA								
Billings AP	45	-10	7150	91	64	0	0	31 H
Bozeman	45	-14		87	60	0	0	32 H
Butte AP	46	-17	9730	83	56	0	0	35 H
Cut Bank AP	48	-20	9033	85	61	0	0	35 H
Glasgow AP (S)	48	-18	9000	89	63	0	0	29 H
Glendive	47	-13		92	64	0	0	29 H
Great Falls AP (S)	47	-15	7670	88	60	0	0	28 H
Havre	48	-11	8880	90	64	0	0	33 H
Helena AP	46	-16	8190	88	60	0	0	32 H
Kalispell AP	48	-7	8150	87	61	0	0	34 H
Lewiston AP	47	-16	8586	87	61	0	0	30 H
Livingston AP	45	-14		87	60	0	0	32 H

AP - Airport
CO - City Office
S - Solar Data Available

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Miles City AP	46	15	7810	95	66	0	0	30 H
Missoula AP	46	-6	8000	88	61	0	0	36 H
NEBRASKA								
Beatrice	40	-2		95	74	21	28	24 M
Chadron AP	42	-3	7100	94	65	0	0	30 H
Columbus	41	-2		95	73	16	23	M
Fremont	41	-2		95	74	21	28	22 M
Grand Island AP	41	-3	6440	94	71	6	13	28 H
Hastings	40	-3	6070	94	71	6	13	27 H
Kearney	40	-4		93	70	2	9	28 H
Lincoln CO (S)	40	-2	6050	95	74	21	28	24 M
McCook	40	-2		95	69	0	0	28 H
Norfolk	42	-4	7010	93	74	25	32	30 H
North Platte AP (S)	41	-4	6680	94	69	0	2	28 H
Omaha AP	41	-3	6290	91	75	35	42	22 M
Scottsbluff AP	41	-3	6670	92	65	0	0	31 H
Sidney AP	41	-3	7030	92	65	0	0	31 H
NEVADA								
Carson City	39	9	5753	91	59	0	0	42 H
Elko AP	40	-2	7430	92	59	0	0	42 H
Ely AP (S)	39	-4	7710	87	56	0	0	39 H
Las Vegas AP (S)	36	28	2610	106	65	0	0	30 H
Lovelock AP	40	12		96	63	0	0	42 H
Reno AP (S)	39	10	6150	92	60	0	0	45 H
Reno CO	39	11		93	60	0	0	45 H
Tonapah AP	38	10	5900	92	59	0	0	40 H
Winnemucca AP	40	3	6760	94	60	0	0	42 H
NEW HAMPSHIRE								
Berlin	44	-9	8270	84	69	11	18	22 M
Claremont	43	-4	7850	86	70	14	21	24 M
Concord AP	43	-3	7360	87	70	12	19	26 H
Keene	43	-7	7460	87	70	12	19	24 M
Laconia	43	-5	7560	86	70	14	21	25 M
Manchester, Grenier AFB	43	-3	7100	88	71	16	23	24 M
Portsmouth, Pease AFB	43	2	6710	85	71	20	27	22 M
NEW JERSEY								
Atlantic City CO	39	13	4810	89	74	31	38	18 M
Long Branch	40	13		90	73	24	31	18 M
Newark AP	40	14	4900	91	73	23	30	20 M
New Brunswick	40	10	5400	89	73	26	33	19 M
Paterson	40	10	5360	91	73	23	30	21 M
Phillipsburg	40	6		89	72	20	27	21 M
Trenton CO	40	14	4980	88	74	33	40	19 M
Vineland	39	11		89	74	31	38	19 M
NEW MEXICO								
Alamogordo								
Holloman AFB	32	19	3240	96	64	0	0	30 H
Albuquerque AP (S)	35	16	4250	94	61	0	0	27 H
Artesia	32	19		100	67	0	0	30 H
Carlsbad AP	32	19	2835	100	67	0	0	28 H
Clovis AP	34	13	4200	93	65	0	0	28 H
Farmington AP	36	6	5713	93	62	0	0	30 H
Gallup	35	5		89	58	0	0	32 H
Grants	35	4		88	58	0	0	32 H
Hobbs AP	32	18		99	66	0	0	29 H
Las Cruces	32	20	3194	96	64	0	0	30 H
Los Alamos	35	9		87	60	0	0	32 H
Raton AP	36	1	6228	89	60	0	0	34 H
Roswell, Walker AFB	33	18	3680	98	66	0	0	33 H
Santa Fe CO	35	10	6120	88	61	0	0	28 H
Silver City AP	32	10	3705	94	60	0	0	30 H
Socorro AP	34	17		95	62	0	0	30 H
Tucumcari AP	35	13	4047	97	66	0	0	28 H
NEW YORK								
Albany AP (S)	42	-1	6900	88	72	22	29	23 M
Albany CO	42	1		88	72	22	29	20 M
Auburn	43	2		87	71	18	25	22 M

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Batavia	43	5		87	71	18	25	22 M
Binghamton AP	42	1	7340	83	69	13	20	20 M
Buffalo AP	43	6	6960	85	70	15	22	21 M
Cortland	42	0		85	71	20	27	23 M
Dunkirk	42	9	6851	85	72	26	33	18 M
Elmira AP	42	1		86	71	19	26	24 M
Geneva (S)	42	2		87	71	18	20	22 M
Glen Falls	42	-5	7270	85	71	20	27	23 M
Gloversville	43	-2		86	71	19	26	23 M
Hornell	42	0		85	70	15	22	24 M
Ithaca (S)	42	0	7052	85	71	20	27	24 M
Jamestown	42	3	6849	86	70	14	21	20 M
Kingston	42	2		88	72	22	29	22 M
Lockport	43	7	6724	86	72	25	32	21 M
Massena AP	45	-8		83	69	13	20	20 M
Newburg-Stewart AFB	41	4	6336	88	72	22	29	21 M
NYC-Central Park (S)	40	15	4880	89	73	26	33	17 M
NYC-Kennedy AP	40	15	5219	87	72	23	30	16 M
NYC-La Guardia AP	40	15	4811	89	73	26	33	16 M
Niagra Falls AP	43	7	6688	86	72	25	32	20 M
Olean	42	2		84	71	22	29	23 M
Oneonta	42	-4		83	69	13	20	24 M
Oswego CO	43	7	6792	83	71	24	31	20 M
Plattsburg AFB	44	-8	8044	83	69	13	20	22 M
Poughkeepsie	41	6	5820	89	74	31	38	21 M
Rochester AP	43	5	6760	88	71	16	23	22 M
Rome-Griffiss AFB	43	-5	7331	85	70	15	22	22 M
Schenectady (S)	42	1	6650	87	72	23	30	22 M
Suffolk County AFB	40	10	5951	83	71	24	31	16 M
Syracuse AP	43	2	6720	87	71	18	25	20 M
Utica	43	-6	7200	85	71	20	27	22 M
Watertown	44	-6	7300	83	71	24	31	20 M
NORTH CAROLINA								
Asheville AP	35	14	4130	87	72	23	30	21 M
Charlotte AP	35	22	3200	93	74	25	32	20 M
Durham	36	20		92	75	33	40	20 M
Elizabeth City AP	36	19	3207	91	77	47	54	18 M
Fayetteville, Pope AFB	35	20	3080	92	76	39	46	20 M
Goldsboro, Seymour-								
Johnson AFB	35	21	3124	91	76	41	48	18 M
Greensboro AP (S)	36	18	3810	91	73	23	30	21 M
Greenville	35	21		91	76	41	48	19 M
Henderson	36	15		92	76	39	46	20 M
Hickory	35	18		90	72	19	26	21 M
Jacksonville	34	24		90	78	55	62	18 M
Lumberton	34	21		92	76	39	46	20 M
New Bern AP	35	24		90	78	55	62	18 M
Raleigh/Durham AP (S)	35	20	3440	92	75	33	40	20 M
Rocky Mount	36	21		91	76	41	48	19 M
Wilmington AP	34	26	2380	91	78	53	60	18 M
Winston-Salem AP	36	20	3650	91	73	23	30	20 M
NORTH DAKOTA								
Bismark AP (S)	46	-19	8960	91	68	0	2	27 H
Devil's Lake	48	-21	9901	88	68	0	7	25 M
Dickinson AP	46	-17	8942	90	66	0	0	25 M
Fargo AP	46	-18	9250	89	71	15	22	25 M
Grands Forks AP	48	-22	9930	87	70	12	19	25 M
Jamestown AP	47	-18		90	69	2	9	26 H
Minot AP	48	-20	9610	89	67	0	0	25 M
Williston	48	-21	9180	88	67	0	1	25 M
OHIO								
Akron-Canton AP	41	6	6140	86	71	19	26	21 M
Ashtabula	41	9		85	72	26	33	18 M
Athens	39	6		92	74	27	34	22 M
Bowling Green	41	2		89	73	26	33	23 M
Cambridge	40	7		90	74	30	37	23 M
Chillicothe	39	6		92	74	27	34	22 M
Cincinnati CO	39	6	4830	90	72	19	26	21 M
Cleveland AP (S)	41	5	6200	88	72	22	29	22 M
Columbus AP (S)	40	5	5670	90	73	24	31	24 M
Dayton AP	39	4	5620	89	72	20	27	20 M
Defiance	41	4		91	73	23	30	24 M

AP - Airport
CO - City Office
S - Solar Data Available

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Findlay AP	41	3		90	73	24	31	24 M
Fremont	41	1		88	73	28	35	24 M
Hamilton	39	5		90	72	19	26	22 M
Lancaster	39	5		91	73	23	30	23 M
Lima	40	4	5870	91	73	23	30	24 M
Mansfield AP	40	5	6403	87	72	23	30	22 M
Marion	40	5		91	73	23	30	23 M
Middletown	39	5		90	72	19	26	22 M
Newark	40	5	5655	92	73	21	28	23 M
Norwalk	41	1		88	73	28	35	22 M
Portsmouth	38	10	4410	92	74	27	34	22 M
Sandusky CO	41	6	5796	91	72	17	24	21 M
Springfield	40	3	4284	89	73	26	33	21 M
Stubenville	40	5		86	71	19	26	22 M
Toledo AP	41	1	6430	88	73	28	35	25 M
Warren	41	5		87	71	18	25	23 M
Wooster	40	6		86	71	19	26	22 M
Youngstown AP	41	4	6370	86	71	19	26	23 M
Zanesville AP	40	7		90	74	30	37	23 M
OKLAHOMA								
Ada	34	14		97	74	19	26	23 M
Altus AFB	34	16	3190	100	73	7	14	25 M
Ardmore	34	17	3060	98	74	17	24	23 M
Bartlesville	36	10		98	74	17	24	23 M
Chickasha	35	14		98	74	17	24	24 M
Enid-Vance AFB	36	13	3971	100	74	13	20	24 M
Lawton AP	34	16	3250	99	74	15	22	24 M
McAlester	34	19	3255	96	74	20	27	23 M
Muskogee AP	35	15		98	75	23	30	23 M
Norman	35	13	3247	96	74	20	27	24 M
Oklahoma City AP	35	13	3700	97	74	19	26	23 M
Ponca City	36	9	3850	97	74	19	26	24 M
Seminole	35	15		96	74	20	27	23 M
Stillwater (S)	36	13		96	74	20	27	24 M
Tulsa AP	36	13	3730	98	75	23	30	22 M
Woodward	36	10		97	73	13	20	26 H
OREGON								
Albany	44	22		89	66	0	0	31 H
Astoria AP (S)	46	29	5190	71	62	0	4	16 M
Baker AP	44	6		89	61	0	0	30 H
Bend	44	4		87	60	0	0	33 H
Corvallis (S)	44	22	4854	89	66	0	0	31 H
Eugene AP	44	22	4740	89	66	0	0	31 H
Grants Pass	42	24	4375	96	68	0	0	33 H
Klamath Falls AP	42	9	6810	87	60	0	0	36 H
Medford AP (S)	42	23	4880	94	67	0	0	35 H
Pendleton AP	45	5	4700	93	64	0	0	29 H
Portland AP	45	23	4635	85	67	0	7	23 M
Portland CO	45	24		86	67	0	4	21 M
Roseburg AP	43	23	4491	90	66	0	0	30 H
Salem AP	45	23	4760	88	66	0	0	31 H
The Dalles	45	19		89	68	3	10	28 H
PENNSYLVANIA								
Allentown AP	40	9	5810	88	72	22	29	22 M
Altoona CO	40	5	6192	87	71	18	25	23 M
Butler	40	6		87	72	23	30	22 M
Chambersburg	40	8	5170	90	74	30	37	23 M
Erie AP	42	9	6540	85	72	26	33	18 M
Harrisburg AP	40	11	5280	91	74	29	36	21 M
Johnstown	40	2	7804	83	70	19	26	23 M
Lancaster	40	8	5560	90	74	30	37	22 M
Meadville	41	4		85	70	15	32	21 M
New Castle	41	7	5800	88	72	22	30	23 M
Philadelphia AP	39	14	5180	90	74	30	37	21 M
Pittsburgh AP	40	5	5950	86	71	19	26	22 M
Pittsburgh CO	40	7		88	71	16	22	19 M
Reading CO	40	13	4960	89	72	20	22	19 M
Scranton/Wilkes-Barre	41	5	6160	87	71	18	25	19 M
State College (S)	40	7	6160	87	71	18	25	23 M
Sunbury	40	7		89	72	20	27	22 M
Uniontown	39	9		88	73	28	35	22 M
Warren	41	4		86	71	19	26	24 M

Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
West Chester	40	13		89	74	31	38	20 M
Williamsport AP	41	7	5950	89	72	20	27	23 M
York	40	12		91	74	29	36	22 M
RHODE ISLAND								
Newport (S)	41	9	5800	85	72	26	33	16 M
Providence AP	41	9	5950	86	72	25	32	19 M
SOUTH CAROLINA								
Anderson	34	23		92	74	27	34	21 M
Charleston AFB (S)	32	27	2070	91	78	53	60	18 M
Charleston CO	32	28	2146	92	78	51	58	13 L
Columbia AP	34	24	2520	95	75	28	35	22 M
Florence AP	34	25	2480	92	77	45	52	21 M
Georgetown	33	26	2228	90	78	55	62	18 M
Greenville AP	34	22	3070	91	74	29	36	21 M
Greenwood	34	22	2890	93	74	25	32	21 M
Orangeburg	33	24		95	75	28	35	20 M
Rock Hill	35	23		94	74	23	30	20 M
Spartanburg AP	35	22		91	74	29	36	20 M
Sumter-Shaw AFB	34	25	2453	92	76	39	46	21 M
SOUTH DAKOTA								
Aberdeen AP	45	-15	8620	91	72	17	24	27 H
Brookings	44	-13		92	72	15	22	25 M
Huron AP	44	-14	8220	93	72	13	20	28 H
Mitchel	43	-10		93	71	7	14	28 H
Pierre AP	44	-10	7550	95	71	4	11	29 H
Rapid City AP (S)	44	-7	7370	92	65	0	0	28 H
Sioux Falls AP	43	-11	7840	91	72	17	24	24 M
Watertown AP	45	-15	8390	91	72	17	24	26 H
Yankton	43	-7		91	72	17	24	25 M
TENNESSEE								
Athens	33	18		92	73	21	28	22 M
Bristol-Tri City AP	36	14	4143	89	72	20	27	22 M
Chattanooga AP	35	18	3380	93	74	25	32	22 M
Clarksville	36	12		93	74	25	32	21 M
Columbia	35	15		94	74	23	30	21 M
Dyersburg	36	15		94	77	42	49	21 M
Greenville	35	16		90	72	19	26	22 M
Jackson AP	35	16	3350	95	75	28	35	21 M
Knoxville AP	35	19	3510	92	73	21	28	21 M
Memphis AP	35	18	3210	95	76	33	48	21 M
Murfreesboro	35	14		94	74	23	30	22 M
Nashville AP (S)	36	14	3610	94	74	23	30	21 M
Tullahoma	35	13	3577	93	73	19	26	22 M
TEXAS								
Abilene AP	32	20	2620	99	71	0	5	22 M
Alice AP	27	34		98	77	35	42	20 M
Amarillo AP	35	11	4140	95	67	33	40	26 H
Austin AP	30	28	1720	98	74	23	30	22 M
Bay City	29	33		94	77	42	49	16 M
Beaumont	30	31	1370	93	78	50	57	19 M
Beeville	28	33	1189	97	77	37	44	18 M
Big Springs AP (S)	32	20		97	69	0	0	26 H
Brownsville AP (S)	25	39	600	93	77	44	51	18 M
Brownwood	31	22	2437	99	73	9	16	22 M
Bryan AP	30	29	1640	96	76	32	39	20 M
Corpus Christi AP	27	35	930	94	78	49	56	19 M
Corsicana	32	25		98	75	23	30	21 M
Dallas AP	32	22	2320	100	75	20	27	20 M
Del Rio, Laughlin AFB	29	31	1520	98	73	11	18	24 M
Denton	33	22		99	74	15	22	22 M
Eagle Pass	28	32	1423	99	73	9	16	24 M
El Paso AP (S)	31	24	2680	98	64	0	0	27 H
Fort Worth AP (S)	32	22	2390	99	74	15	22	22 M
Galveston AP	29	36	1270	89	79	63	70	10 L
Greenville	33	22		99	74	15	22	21 M
Harlingen	26	39	693	94	77	42	49	19 M
Houston AP	29	32	1410	94	77	42	49	18 M
Houston CO	29	33		95	77	40	47	18 M
Huntsville	30	27		98	75	23	30	20 M
Killeen-Gray AFB	31	25		97	73	13	20	22 M

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Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER		SUMMER				
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Lamesa	32	17		96	69	0	0	26 H
Laredo AFB	27	36	800	101	73	6	13	23 M
Longview	32	24		97	76	30	37	20 M
Lubbock AP	33	15	3570	96	69	0	0	26 H
Lufkin AP	31	29	1940	97	76	30	37	20 M
McAllen	26	39		95	77	40	47	21 M
Midland AP (S)	32	21	2600	98	69	0	0	26 H
Mineral Wells AP	32	22		99	74	15	22	22 M
Palestine CO	31	27		98	76	29	36	20 M
Pampa	35	12		96	67	0	0	26 H
Pecos	31	21		98	69	0	0	27 H
Plainview	34	13		96	68	0	0	26 H
Port Arthur AP	30	31	1447	93	78	50	57	19 M
San Angelo, Goodfellow AFB	31	22	2220	99	71	0	5	24 M
San Antonio AP (S)	29	30	1560	97	73	13	20	19 M
Sherman-Perrin AFB	33	20	2837	98	75	23	30	22 M
Snyder	32	18		98	70	0	1	26 M
Temple	31	27		99	74	15	22	22 M
Tyler AP	32	24		97	76	30	37	21 M
Vernon	34	17		100	73	7	14	24 M
Victoria AP	28	32	1160	96	77	39	46	18 M
Waco AP	31	26	2040	99	75	21	28	22 M
Wichita Falls AP	34	18	2900	101	73	6	13	24 M
UTAH								
Cedar City AP	37	5	5680	91	60	0	0	32 H
Logan	41	2	6750	91	61	0	0	33 H
Moab	38	11		98	60	0	0	30 H
Ogden AP	41	5	6012	91	61	0	0	33 H
Price	39	5		91	60	0	0	33 H
Provo	40	6	5720	96	62	0	0	32 H
Richfield	38	5		91	60	0	0	34 H
St. George CO	37	21		101	65	0	0	33 H
Salt Lake City AP (S)	40	8	5990	95	62	0	0	32 H
Vernal AP	40	0		89	60	0	0	32 H
VERMONT								
Barre	44	-11		81	69	16	23	23 M
Burlington AP (S)	44	-7	8030	85	70	15	22	23 M
Rutland	43	-8	7440	84	70	16	23	23 M
VIRGINIA								
Charlottesville	38	18	4220	91	74	29	36	23 M
Danville AP	36	16	3510	92	73	21	28	21 M
Fredericksburg	38	14		93	75	31	38	21 M
Harrisonburg	38	16		91	72	17	24	23 M
Lynchburg AP	37	16	4166	90	74	30	37	21 M
Norfolk AP	36	22	3440	91	76	41	48	18 M
Petersburg	37	17		92	76	39	46	20 M
Richmond AP	37	17	3910	92	76	39	46	21 M
Roanoke AP	37	16	4150	91	72	17	24	23 M
Staunton	38	16	4307	91	72	17	24	23 M
Winchester	39	10	4780	90	74	30	37	21 M
WASHINGTON								
Aberdeen	47	28	5316	77	62	0	0	16 M
Bellingham AP	48	15	5420	77	65	2	9	19 M
Bremerton	47	25	5432	78	64	0	3	20 M
Ellensburg AP	47	6		91	64	0	0	34 H
Everett-Paine AFB	47	25	5940	76	64	0	7	20 M
Kennewick	46	11		96	67	0	0	30 H
Longview	46	24	5064	85	67	0	7	30 H
Moses Lake, Larson AFB	47	7		94	65	0	0	32 H
Olympia AP	47	22	5236	83	65	0	0	32 H
Port Angeles	48	27		69	61	0	3	18 M
Seattle-Boeing Fld	47	26		81	66	1	8	24 M
Seattle CO (S)	47	27	4424	82	66	0	7	19 M
Seattle-Tacoma AP (S)	47	26	5190	80	64	0	0	22 M
Spokane AP (S)	47	2	6770	90	63	0	0	28 M
Tacoma-McChord AFB	47	24	5287	82	65	0	2	22 M
Walla Walla AP	46	7	4800	94	66	0	0	27 H
Wenatchee	47	11		96	66	0	0	32 H
Yakima AP	46	5	5950	93	65	0	0	36 H

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Table 1 (CONTINUED)

		WINTER		SUMMER				
Location	Latitude Degrees	97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
WEST VIRGINIA								
Beckley	37	4	5615	81	69	16	23	22 M
Bluefield AP	37	4	5000	81	69	16	23	22 M
Charleston AP	38	11	4510	90	73	24	31	20 M
Clarksburg	39	10	4590	90	73	24	31	21 M
Elkins AP	38	6	5680	84	70	16	23	22 M
Huntington CO	38	10	4340	91	74	29	36	22 M
Martinsburg AP	39	10	5231	90	74	30	37	21 M
Morgantown AP	39	8	5100	87	73	29	36	22 H
Parkersburg CO	39	11	4780	90	74	30	37	21 M
Wheeling	40	5	5220	86	71	19	26	21 M
WISCONSIN								
Appleton	44	-9		86	72	25	32	23 M
Ashland	46	-16		82	68	10	17	23 M
Beloit	42	-3		90	75	36	43	24 M
Eau Claire AP	44	-11	7970	89	73	26	33	23 M
Fond du Lac	43	-8		86	72	25	32	23 M
Green Bay AP	44	-9	8100	85	72	26	33	23 M
LaCrosse AP	43	-9	7530	88	73	28	35	22 M
Madison AP (S)	43	-7	7720	88	73	28	35	22 M
Manitowoc	44	-7		86	72	25	32	21 M
Marinette	45	-11		84	71	32	29	20 M
Milwaukee AP	43	-4	7470	87	73	29	36	21 M
Racine	42	-2		88	73	28	35	21 M
Sheboygan	43	-6		86	73	31	38	20 M
Stevens Point	43	-11		89	73	26	33	23 M
Waukesha	43	-5		87	73	29	36	22 M
Wausau AP	44	-12	8490	88	72	22	29	23 M
WYOMING								
Casper AP	42	-5	7510	90	57	0	0	31 H
Cheyene AP	41	-1	7370	86	58	0	0	30 H
Cody AP	44	-13		86	60	0	0	32 H
Evanston	41	-3		84	55	0	0	32 H
Lander AP (S)	42	-11	7870	88	61	0	0	32 H
Laramie AP (S)	41	-6	7560	81	56	0	0	28 H
Newcastle	43	-12		87	63	0	0	30 H
Rawlins	41	-4		83	57	0	0	40 H
Rock Springs AP	41	-3	8430	84	55	0	0	32 H
Sheridan AP	44	-8	7740	91	62	0	0	32 H
Torrington	42	-8		91	62	0	0	30 H

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Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER			SUMMER			Daily Range
		97½ % Design db	Heating D.D. Below 65°F	2½ % Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	
ALBERTA								
Calgary AP	51	-23	9703	81	61	0	0	25 M
Edmonton AP	53	-25	10268	82	65	0	2	23 M
Grande Prairie AP	55	-33	11129	80	63	0	0	23 M
Jasper	52	-26	10112	80	62	0	0	28 H
Lethbridge AP (S)	49	-22	8644	87	63	0	0	28 H
McMurray AP	56	-38	12462	82	65	0	2	26 H
Medicine Hat AP	50	-24	8852	90	65	0	0	28 H
Red Deer AP	52	-26	10302	81	64	0	0	25 M
BRITISH COLUMBIA								
Dawson Creek	55	-33	10800	79	63	0	0	26 H
Fort Nelson AP (S)	58	-40	10874	81	63	0	0	23 M
Kamloops CO	50	-15	6799	91	65	0	0	29 H
Nanaimo (S)	49	20	5554	80	65	0	4	21 M
New Westminster	49	18		81	67	5	12	19 M
Penticton AP	49	4	6522	89	67	0	0	31 H
Prince George AP (S)	53	-28	9755	80	62	0	0	26 H
Prince Rupert CO	54	2	2029	63	57	0	0	12 L
Trail	49	0	6711	89	65	0	0	33 H
Vancouver AP (S)	49	19	5515	77	66	8	15	17 M
Victoria CO	48	23	5579	73	62	0	0	16 M
MANITOBA								
Brandon	49	-27	10828	86	70	14	21	25 M
Churchill AP (S)	58	-39	16728	77	64	0	3	18 M
Dauphin AP	51	-28		84	70	16	23	23 M
Flin Flon	54	-37	12414	81	66	2	9	19 M
Portage la Prairie AP	49	-24	10800	86	72	25	32	22 M
The Pas AP (S)	53	-33	12281	82	67	4	11	20 M
Winnipeg AP (S)	49	-27	10679	86	71	19	26	22 M
NEW BRUNSWICK								
Campbellton CO	48	-14		82	67	4	11	21 M
Chatham AP	47	-10		85	68	5	12	22 M
Edmundston CO	47	-16	9796	83	68	8	15	21 M
Fredericton AP (S)	45	-11	8671	85	69	9	11	23 M
Moncton AP (S)	46	-8	8711	82	69	14	21	23 M
Saint John AP	45	-8	8453	77	65	3	10	19 M
NEWFOUNDLAND								
Corner Brook	48	0	8978	73	63	0	5	17 M
Gander AP	48	-1	9254	79	65	0	7	19 M
Goose Bay AP (S)	53	-24	11887	81	64	0	0	19 M
St. John's AP (S)	47	7	8991	75	65	6	13	18 M
Stephenville AP	48	4	8717	74	64	3	10	14 L
NORTHWEST TERR.								
Fort Smith AP (S)	60	-45		81	64	0	0	24 M
Frobisher AP (S)	83	-41	17876	63	51	0	0	14 L
Inuvik (S)	68	-53		77	60	0	0	21 M
Resolute AP (S)	74	-47	22673	54	46	0	0	10 L
Yellowknife AP	62	-46	15634	77	61	0	0	16 M
NOVA SCOTIA								
Amherst	45	-6	8400	81	68	11	18	21 M
Halifax AP (S)	44	5	7361	76	65	3	10	16 M
Kentville (S)	45	1	7792	83	68	8	15	22 M
New Glasgow	45	-5		79	68	14	21	20 M
Sydney AP	46	3	8049	80	68	12	19	19 M
Truro CO	45	-5	8226	80	69	18	25	22 M
Yarmouth AP	43	9	7340	72	64	6	13	15 L
ONTARIO								
Belleville	44	-7	7709	84	72	28	35	20 M
Chatham	42	3	6503	87	73	29	36	19 M
Cornwall	45	-9	8200	87	72	23	30	21 M
Hamilton	43	1	6821	86	72	25	32	21 M
Kaupuskasing AP (S)	49	-28	11560	83	69	13	20	23 M
Kenora AP	49	-28	10796	82	69	14	21	19 M
Kingston	44	-7	7724	84	72	28	35	20 M
Kitchener	43	-2	7566	85	72	26	33	23 M
London AP	43	0	7349	85	73	32	39	21 M
North Bay AP	46	-18	9677	81	67	6	13	20 M
Oshawa	43	-3	7600	86	72	25	32	20 M

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Table 1 (CONTINUED)

Location	Latitude Degrees	WINTER			SUMMER			
		97½% Design db	Heating D.D. Below 65°F	2½% Design db	Coincident Design wb	Grains Difference 55% RH	Grains Difference 50% RH	Daily Range
Ottawa AP (S)	45	-13	8693	87	71	18	25	21 M
Owen Sound	44	-2	...	82	70	20	27	21 M
Peterborough	44	-9	8309	85	71	20	27	21 M
St. Catharines	43	3	6537	85	72	26	33	20 M
Sarnia	42	3	7061	86	72	25	32	19 M
Sault Ste. Marie AP	46	-13	9500	82	69	14	21	22 M
Sudbury AP	46	-19	9600	83	67	3	10	22 M
Thunder Bay AP	48	-24	10405	83	68	8	15	24 M
Timmins AP	48	-29	11400	84	68	6	13	25 M
Toronto AP (S)	43	-1	6827	87	72	23	30	20 M
Windsor AP	42	4	6579	88	73	28	35	20 M
PRINCE EDWARD ISLAND								
Charlottetown AP (S)	46	-4	8486	78	68	16	23	16 M
Summerside AP	46	-4	8440	79	68	14	21	16 M
QUEBEC								
Bagotville AP	48	-23	...	83	68	8	15	21 M
Chicoutimi	48	-22	10104	83	68	8	15	20 M
Drummondville	45	-14	8700	85	71	20	27	21 M
Granby	45	-14	8400	85	71	20	27	21 M
Hull	45	-14	8700	87	71	18	25	21 M
Megantic AP	45	-16	...	83	70	19	26	20 M
Montreal AP (S)	45	-10	8213	85	72	26	33	17 M
Quebec AP	46	-14	8937	84	70	16	23	20 M
Rimouski	48	-12	9906	79	66	4	11	18 M
St. Jean	45	-11	8500	86	72	25	32	20 M
St. Jerome	45	-13	9285	86	71	19	26	23 M
Sept. Iles AP (S)	50	-21	...	73	61	0	0	17 M
Shawinigan	46	-14	9380	84	70	16	23	21 M
Sherbrooke CO	45	-21	8490	84	71	22	29	20 M
Thetford Mines	46	-14	9815	84	70	16	23	21 M
Trois Rivières	46	-13	9306	85	70	15	22	23 M
Val d'Or AP	48	-27	11169	83	68	8	15	22 M
Valleyfield	45	-10	8300	86	72	25	32	20 M
SASKATCHEWAN								
Estevan AP	49	-25	9950	89	68	0	5	26 H
Moose Jaw AP	50	-25	9894	89	67	0	0	27 H
North Battleford AP	52	-30	11082	85	66	0	0	23 M
Prince Albert AP	53	-35	11630	84	66	0	2	25 M
Regina AP	50	-29	10806	88	68	0	7	26 H
Saskatoon AP (S)	52	-31	10856	86	66	0	0	26 H
Swift Current AP (S)	50	-25	9849	90	66	0	0	25 M
Yorkton AP	51	-30	11362	84	68	6	13	23 M
YUKON TERRITORY								
Whitehorse AP (S)	60	-43	12475	77	58	0	0	22 M

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